

Consider the following polyhedron represented in standard form:

$$\mathcal{P} = \{\mathbf{x} \in \mathbb{R}^4 \mid A\mathbf{x} = \mathbf{b}, \mathbf{x} \geq 0\},$$

with

$$A = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & -1 & 0 & 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Write the polyhedron in canonical form.